CS & IT ENGINEERING

Data Structure and Programming

Tree

DPP- 01 Discussion Notes



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[NAT]

#Q. The number of unlabeled binary trees possible with four nodes is 14.

Catalan No. $C_n = \frac{1}{n+1} {}^{2n}C_n$

$$C_4 = \frac{1}{5} {}^{8}C_4$$

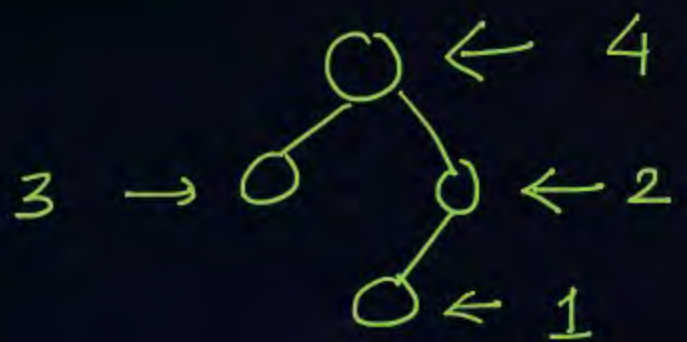
$$= \frac{1}{5} \frac{\lfloor 8}{\lfloor 4 \lfloor 4}$$

$$= \frac{1}{5} \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 14$$

[NAT]

#Q. The number of labelled binary trees possible with the nodes-10, 30, 25, 40 is 336.

No. of unlabelled trees = 14



No. of ways we ~~cant~~ can label

= 4!

No. of labelled trees = 14 × 4!

= 14 × 24 = 336

[NAT]

#Q. The number of binary search trees possible with thenodes-10, 30, 25, 40 is __14__.



No. of Binay Search tree

one labelled tree can filled only oneway Hence

BST = # of unlabelled tres

= 14

[MCQ]

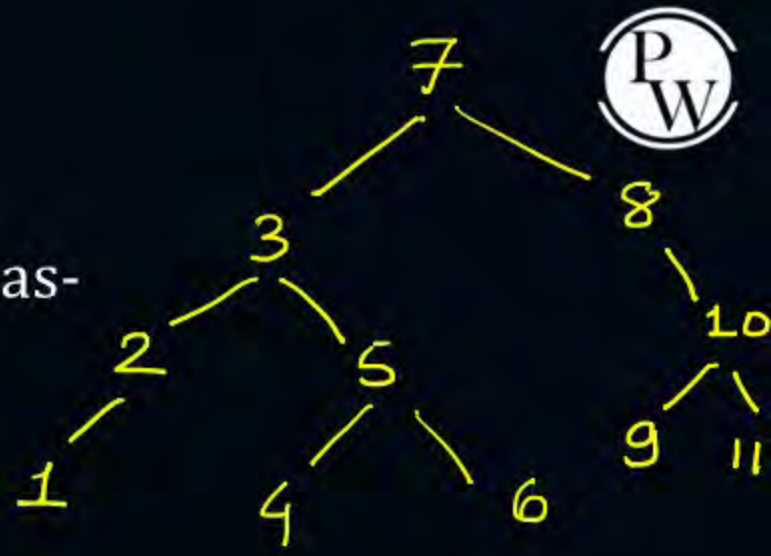
Root Left Right

Left

#Q. The pre-order traversal of a binary search tree is given as-

7, 3, 2, 1, 5, 4, 6, 8, 10, 9, 11

The post-order traversal of the above binary tree is-



A 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

B 1, 2, 4, 6, 5, 3, 9, 11, 10, 8, 7

C 1, 2, 4, 5, 6, 3, 9, 10, 11, 8, 7

D 11, 9, 10, 8, 6, 4, 5, 1, 2, 3, 7

only preorder

postorder : 1, 2, 4, 6, 5, 3

Answer B

[MCQ]



2, 1, 3

1 2 3

#Q. Consider the following two statements:

Statement P: The last elements in the pre-order and in-order traversal of a binary search tree are always same. Incorrect

Statement Q: The last elements in the pre-order and in-order traversal of a binary tree are always same. Incorrect

Which of the following tree is/are CORRECT?



preorder

2 1

Inorder

1, 2

A Both P and Q only

B Neither P nor Q

C Q only

D P only

BST – Binary Tree

[MCQ]

Linked List

tree Node

#Q. Consider the following function:

```
struct treenode{
struct treenode *left;
int data;
struct treenode *right;
};
int func (struct treenode *t){
if(t==NULL) return 1;
else if(t->left==NULL && t->right==NULL)
return 1;
else if
((t → left → data < t->data) && (t → right → data > t->data))
```

tree node pointer is Input.

(1) if t is NULL



Zero

1 element

Continue...

Recursion (BST) Tree structure

```
return func(t->left) && func(t->right);
else
return 0;
}
```

↑ Logical operator && operator

Recursive

Assume t contains the address of the root node of a tree. The function-

A. Returns 1 if the given tree is a Binary Search Tree.

Answer is A

B. Returns 0 if the given tree is a complete binary tree.

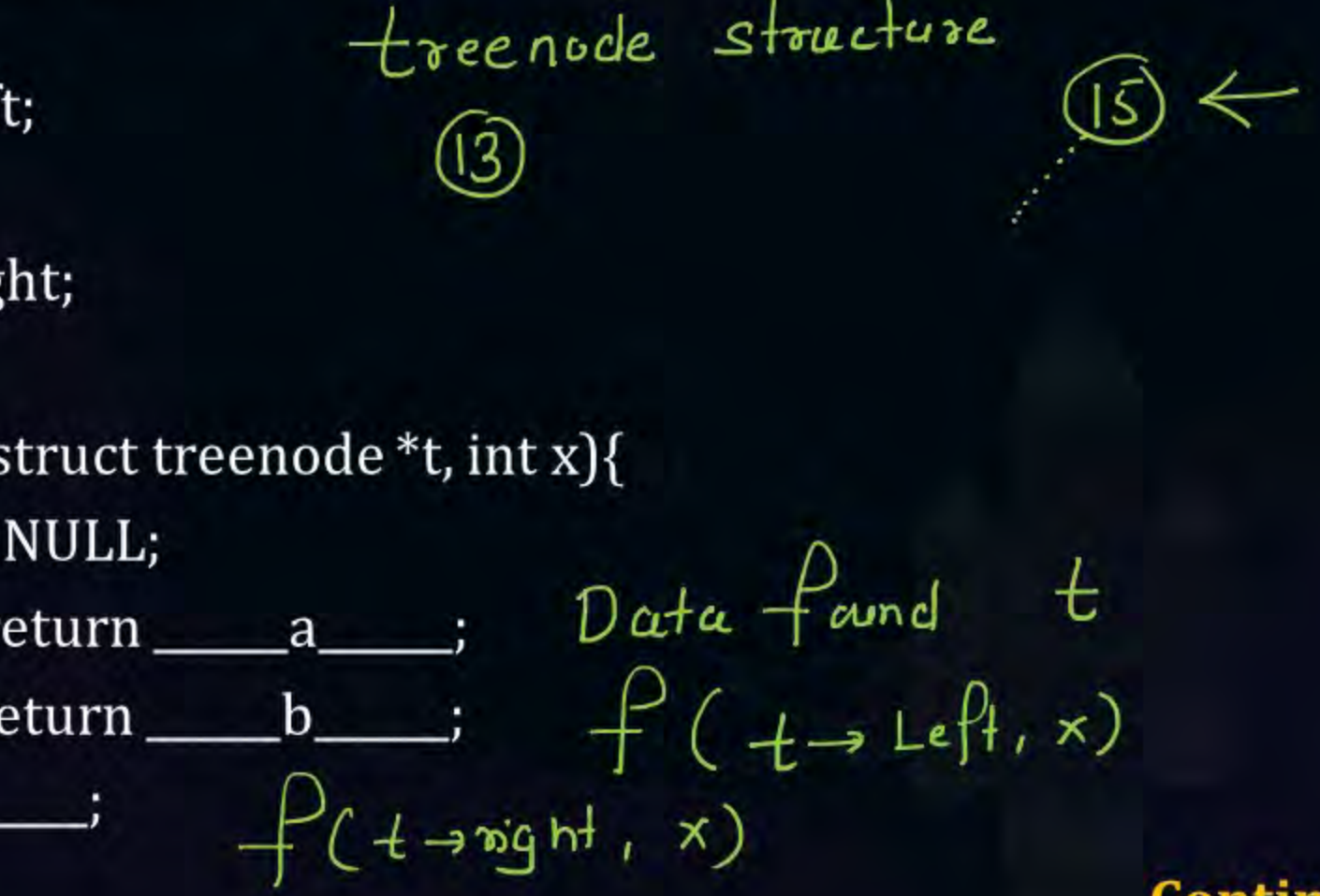
C. Returns 0 if the given tree is a Binary Search Tree.

D. Returns 1 if the given tree is a complete binary tree.

[MCQ]

#Q. Consider the following function:

```
struct treenode{
struct treenode *left;
int data;
struct treenode *right;
};
struct treenode * f(struct treenode *t, int x){
if(t==NULL) return NULL;
elseif(x==t->data) return _____a_____;
else if (x<t->data) return _____b_____;
else return _____c_____;
}
```



Continue...

Searching algorithm

Root Node

[BST]

Assume t contains the address of the root node of a binary search tree. The function finds an element x in the BST and returns the address of the node if found.

Which of the following statement(s) is/are CORRECT?

Address of Node

A a: NULL ; b: f(t->left, x) ; c: f(t->right, x)

B a: t ; b: f(t->right, x) ; c: f(t->left, x)

D a: NULL ; b: f(t->right, x) ; c: f(t->left, x)

C

Answer- D

C a: t ; b: f(t->left, x) ; c: f(t->right, x) ✓

D

THANK - YOU